

Sentiment Analysis Extension

<https://github.com/aiyshwary/Video-based-and-audio-based-sentiment-analysis>

Python

Hugging Face

NLP

RapidMiner

Audio Processing

OVERVIEW

A multimodal sentiment analysis project that independently extracts and analyses sentiment from both the video (visual) and audio streams of a recording, combining insights from two separate NLP and signal processing pipelines.

IMPACT

Built a dual-stream sentiment analysis pipeline that separates video and audio from a recording and independently classifies sentiment from each modality, enabling richer multimodal emotional understanding.

KEY DETAILS

- Extracts audio track from video files and applies speech-to-text followed by NLP-based sentiment classification.
- Processes video frames independently to infer visual sentiment cues from facial expressions and scene context.
- Outputs positive, neutral, or negative sentiment labels with confidence scores per modality.
- Designed for use cases including customer feedback analysis, social media monitoring, and presentation review.
- Implemented using Hugging Face pretrained models and RapidMiner for pipeline orchestration.